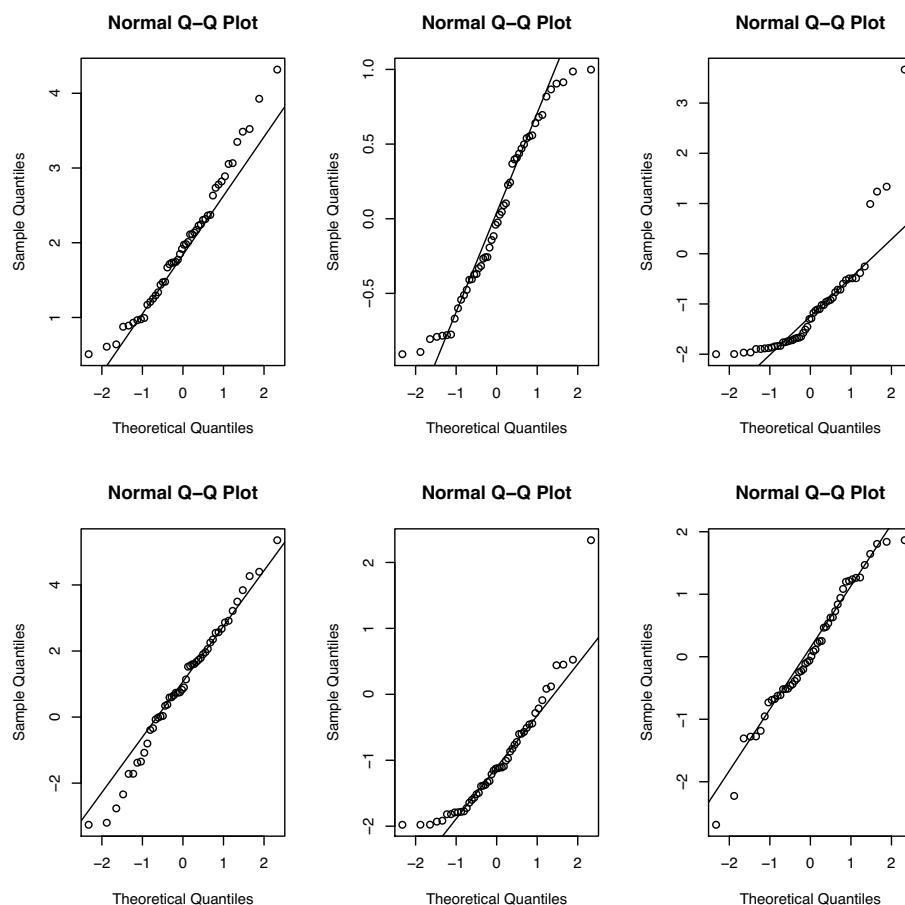


Problem Set 5: QQ Plots and Point Estimates (April 5 – 18, 18:00)

Problem 1 (QQ plots)

The following pictures show normal QQ plots of six different samples of size 50 simulated from the distributions $\mathcal{N}(0, 1)$, $\mathcal{N}(2, 1)$, $\mathcal{N}(1, 2^2)$, $\text{Unif}(-1, 1)$ and twice the distribution of $Y := Z - 2$, where $Z \sim \text{Exp}(1)$. Determine which picture corresponds to which distribution and explain your answer.



Problem 2 (Wind Speeds)

To make more accurate predictions regarding the flight times of their airplanes, the airline SwitzAir wants to use a Rayleigh distribution for measuring wind speeds. This distribution has the cumulative distribution function:

$$F(x) = \begin{cases} 0 & \text{for } x < 0 \\ 1 - \exp\left(-\frac{x^2}{2\sigma^2}\right) & \text{for } x \geq 0 \end{cases}$$

where $\sigma > 0$ is an unknown parameter. The research department assigns us the task of estimating the unknown parameter σ of the distribution based on n observed data points.

Hint: The Rayleigh distribution was discussed in Problem Set 3. You can use that a Rayleigh(σ)-distributed random variable X has expectation $\mathbb{E}[X] = \sqrt{\frac{\pi}{2}}\sigma$ and variance $\text{Var}(X) = \frac{4-\pi}{2}\sigma^2$.

- a) Does this distribution have a density? If yes, can you derive it?
- b) Determine the likelihood and the log-likelihood functions based on the n independent, identically distributed observations $x_1, \dots, x_n$ of a random variable with a Rayleigh distribution. Calculate the maximum likelihood estimator for the unknown parameter σ .

Since SwissAir does not have instruments for measuring wind speeds, they hire a meteorological institute to conduct 100 wind speed measurements. The meteorological institute provides the following empirical statistics: mean $\bar{x} = 2.01$, standard deviation $s_x = 0.51$, median $m = 3.12$, and interquartile range $q_{0.75} - q_{0.25} = 0.89$.

- c) Use this data to compute the maximum likelihood estimate of σ .
- d) Calculate the method of moments estimator based on the first moment. What value does the estimator take for the data collected by the meteorological institute?

Problem 3 (Estimating the Parameter of a Geometric Distribution)

We consider the geometric distribution. As a reminder: this is a discrete distribution with probability mass function

$$\mathbb{P}[X = x] = (1 - p)^{x-1}p, \quad x = 1, 2, 3, \dots$$

where $0 < p < 1$ is the probability of success. A geometrically distributed random variable describes the number of trials until the first success in independent Bernoulli trials with success probability p . We want to estimate the parameter p from a sample.

- a) Determine the likelihood function based on n independent identically distributed observations $x_1, \dots, x_n$ of a geometrically distributed random variable.
- b) Determine the corresponding maximum likelihood estimator for p .
- c) Determine the method of moments estimator for p (again based on n independent observations $x_1, \dots, x_n$). Compare with the maximum likelihood estimator.
- d) Assume you have only a single observation x of a geometrically distributed random variable. We can interpret the experiment as follows: x independent Bernoulli trials were conducted and exactly one success was observed. Write down the likelihood function for this experiment: what is the difference compared to the likelihood function found in a) (for $n = 1$)? Why do you obtain the same maximum likelihood estimator?

Problem 4 (Estimating the Scale Parameter of a Pareto Distribution)

Let $X_1, \dots, X_n$ be i.i.d. Pareto random variables distributed according to the following density function for an unknown parameter $\theta > 0$:

$$f(x; \theta) = \begin{cases} 3\theta^3 x^{-4} & \text{for } x \geq \theta \\ 0 & \text{for } x < \theta \end{cases}$$

- a) Calculate the method of moments estimator for θ and prove that this estimator is unbiased.
- b) Calculate the maximum likelihood estimator for θ and prove that this estimator is biased.
- c) Define, for an estimator $\hat{\theta}$ of the true parameter θ , the *Mean Squared Error* as

$$\text{MSE}(\hat{\theta}, \theta) := \mathbb{E}[(\hat{\theta} - \theta)^2]$$

- 1. Prove that $\text{MSE}(\hat{\theta}, \theta) = \text{Bias}(\hat{\theta}, \theta)^2 + \text{Var}(\hat{\theta})$, where $\text{Bias}(\hat{\theta}, \theta) = \mathbb{E}[\hat{\theta} - \theta]$.
- 2. Which of the estimators obtained in a) and b) has a lower Mean Squared Error?

Problem 5 (Phone Study)

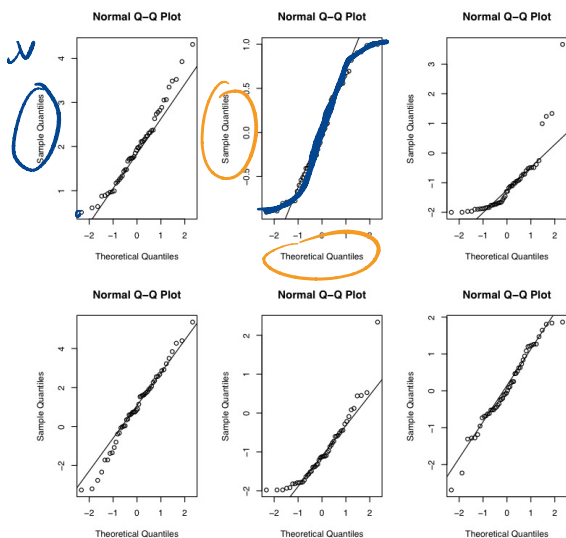
For a study, randomly selected individuals are asked whether they have used drugs in the past year. When asked directly, such a question is often answered with “no” since it is the socially more accepted answer.

To avoid this, one can proceed as follows instead: ask the person to flip a coin and remember the result - but to keep it to themselves. The person should now, if the coin shows heads, answer “yes”, and if the coin shows tails, answer the question.

- a) Why is it more likely that people will tell the truth this way?
- b) Let us take a random sample of 100 individuals from the population and survey them about drug use in the past year using the method suggested above. Assuming 65 individuals respond “yes”, calculate the moment estimator for the proportion of the population that has used drugs in the past year.

Problem 1 (QQ plots)

The following pictures show normal QQ plots of six different samples of size 50 simulated from the distributions $\mathcal{N}(0, 1)$, $\mathcal{N}(2, 1)$, $\mathcal{N}(1, 2^2)$, $\text{Unif}(-1, 1)$ and twice the distribution of $Y := Z - 2$, where $Z \sim \text{Exp}(1)$. Determine which picture corresponds to which distribution and explain your answer.



- QQ-plots are

Problem 3 (Estimating the Parameter of a Geometric Distribution)

We consider the geometric distribution. As a reminder: this is a discrete distribution with probability mass function

$$\mathbb{P}[X = x] = (1 - p)^{x-1}p, \quad x = 1, 2, 3, \dots$$

where $0 < p < 1$ is the probability of success. A geometrically distributed random variable describes the number of trials until the first success in independent Bernoulli trials with success probability p . We want to estimate the parameter p from a sample.

- Determine the likelihood function based on n independent identically distributed observations $x_1, \dots, x_n$ of a geometrically distributed random variable.
- Determine the corresponding maximum likelihood estimator for p .
- Determine the method of moments estimator for p (again based on n independent observations $x_1, \dots, x_n$). Compare with the maximum likelihood estimator.
- Assume you have only a single observation x of a geometrically distributed random variable. We can interpret the experiment as follows: x independent Bernoulli trials were conducted and exactly one success was observed. Write down the likelihood function for this experiment: what is the difference compared to the likelihood function found in a) (for $n = 1$)? Why do you obtain the same maximum likelihood estimator?

MLE is a function of the parameter with observations as parameters

Problem 4 (Estimating the Scale Parameter of a Pareto Distribution)

Let $X_1, \dots, X_n$ be i.i.d. Pareto random variables distributed according to the following density function for an unknown parameter $\theta > 0$:

$$f(x; \theta) = \begin{cases} 3\theta^3 x^{-4} & \text{for } x \geq \theta \\ 0 & \text{for } x < \theta \end{cases}$$

- Calculate the method of moments estimator for θ and prove that this estimator is unbiased.
- Calculate the maximum likelihood estimator for θ and prove that this estimator is biased.
- Define, for an estimator $\hat{\theta}$ of the true parameter θ , the Mean Squared Error as

$$\text{MSE}(\hat{\theta}, \theta) := \mathbb{E}[(\hat{\theta} - \theta)^2]$$

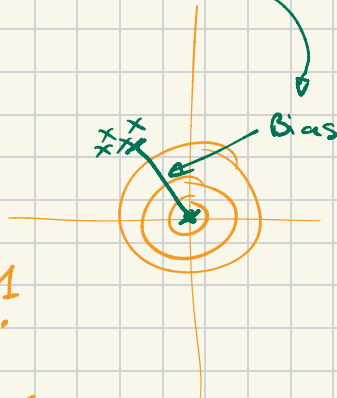
- Prove that $\text{MSE}(\hat{\theta}, \theta) = \text{Bias}(\hat{\theta}, \theta)^2 + \text{Var}(\hat{\theta})$, where $\text{Bias}(\hat{\theta}, \theta) = \mathbb{E}[\hat{\theta} - \theta]$.
- Which of the estimators obtained in a) and b) has a lower Mean Squared Error?

$$E(x) = \frac{9}{7} \theta \quad \hat{\theta} = \frac{7}{9} \hat{\mu} = \frac{7}{9} \left(\sum_{i=1}^n x_i \right) \frac{1}{n}$$

$$E(\hat{\theta}) = \frac{7}{9} \sum_{i=1}^n E(x_i) = \frac{7}{9} \cdot \frac{9}{7} \theta = \theta$$

Bias vs Var. :

Accuracy vs precision



$$\begin{aligned} [&] \Rightarrow \mu_1 \\ [&] \Rightarrow \mu_2 \\ [&] \Rightarrow \mu_3 \end{aligned}$$

Problem 5 (Phone Study)

For a study, randomly selected individuals are asked whether they have used drugs in the past year. When asked directly, such a question is often answered with "no" since it is the socially more accepted answer.

To avoid this, one can proceed as follows instead: ask the person to flip a coin and remember the result - but to keep it to themselves. The person should now, if the coin shows heads, answer "yes", and if the coin shows tails, answer the question.

- Why is it more likely that people will tell the truth this way?
- Let us take a random sample of 100 individuals from the population and survey them about drug use in the past year using the method suggested above. Assuming 65 individuals respond "yes", calculate the moment estimator for the proportion of the population that has used drugs in the past year.

a) There's no way of knowing if you had heads or tails

